Ericsson Response



Ericsson Sustainability
Ambition and strategic areas

To be a responsible and relevant driver of positive change in society

Responsible Business

Environmental sustainability

Digital inclusion

Ericsson Response

THE GLOBAL GOALS For Sustainable Development

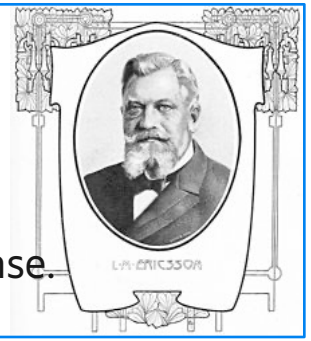
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

17 PARTNERSHIPS FOR THE GOALS

Ericsson Response in brief

Motivation

- In 1876 Lars Magnus Ericsson founded the company on the premise that communication as a basic human need and should be available for all.
- In 2000, a group of employees understood that technology and skills were extremely powerful in humanitarian emergencies and founded Ericsson Response.



Mission

When called by our partners in times of need, Ericsson Response uses its volunteers, technology and competence to support humanitarian relief operations.

50+

Different relief efforts in 40 countries

Since **2000**

Our Principles

- Non-profit
- Non-commercial
- Non-political

Three key elements

1

Volunteers
140+ trained volunteers are ready to support, either by deploying emergency telecoms equipment in the field or working back home, using their expertise to respond to crises as effectively as possible.

2

Technology
The Ericsson Response WiFi solution is our most deployed technology in emergencies, providing connectivity to the humanitarian community and people in need, allowing access to the internet.

In addition, Mobile Technology Solutions, like 2G, 3G and 4G equipment is on stock – ready to deploy.

3

Partners
Ericsson Response supports primarily the UN partner networks UNHCR Refugee Emergency Telcom Sector (RETS) and the WFP-led Emergency Telecom Cluster
Both are mandated to provide services to humanitarian responders in large scale emergencies.

Global Ericsson Program:

140+
Volunteers

30
Countries

29
Languages

Ericsson Response Telco Solutions













WIDER: WiFi in Disaster and Emergency Response



Edge Router

- Modem
- Traffic mgmt.
- Splash page



Cloud AAA

- User mgmt.
  Voucher, Single-Sign-on, registration
- AAA
  Accounting, Authorization, Authentication



Indoor
Access Point



Outdoor
Access Point

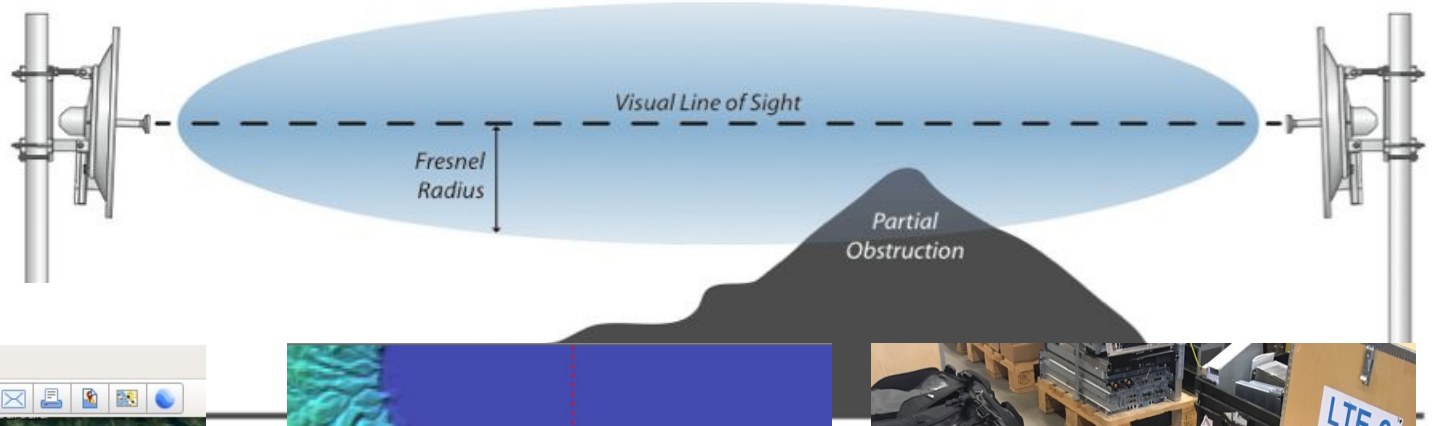


Panel Antenna

Access Control and Cyber security

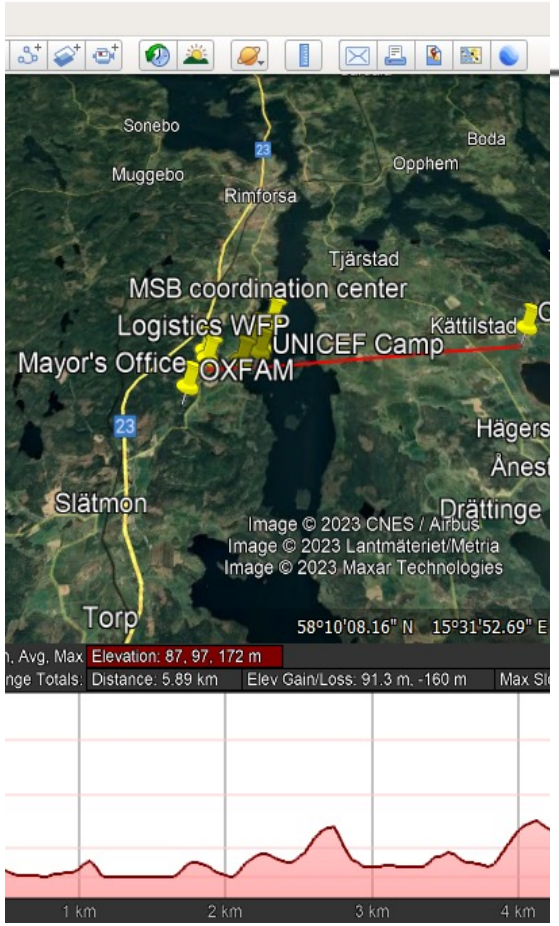
WiFi Network for links and access

The problems we face

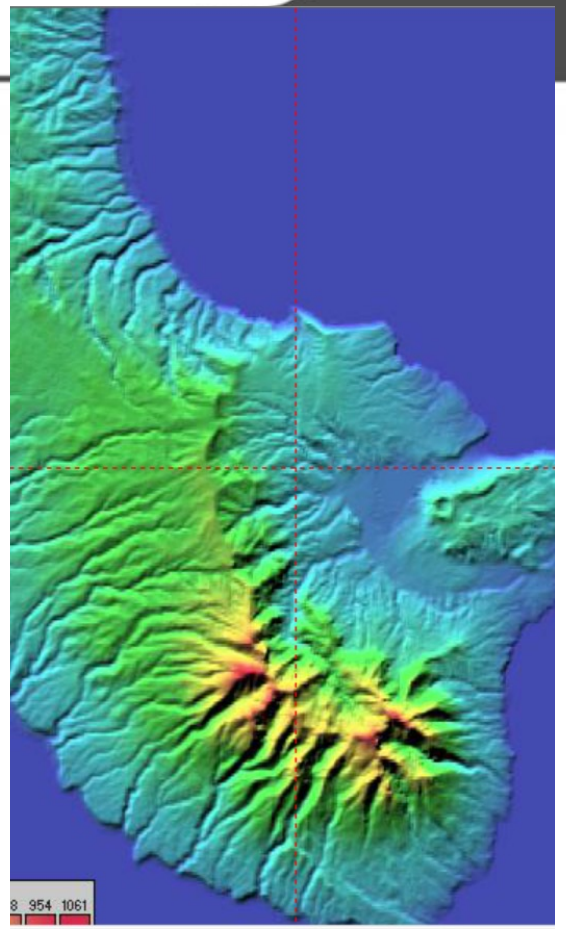


- What does it look like on the ground? what is available? (can we buy stuff?) transportation? (would we need to carry everything?) Security? etc. etc.
- What locations need to be connected, LOS between the different locations, power, etc.
- Find a good location to setup VSAT LOS to the sky, stable power, security

Preparation before and on the ground: we use tools like Google Earth or Radio Mobile (to help us find LOS for links). But in the end you will have to do a site survey on the ground. Since the tools tend to miss Buildings, Trees, etc.



Google Earth



Radio Mobile Tana Island, Vanuatu



Backpacks (23kg)

Evolving….

- Flying Wi-Fi Access Point's for emergency responder

AI Drone carried Wi-Fi antenna (Access Point) For us Areas ex. by (Ericsson Response)

Equip/develop small Drone/(s) carrying a Wi-Fi Access Point (2,4 & 5,8 GHz s standard being used at the moment) that could operate as "flying Access Poi fixed "optimal" position in the sky (as a replacement for broken com-tower) fo or land (hock on to) at a strategic position and power itself and the onboard A power level gets low it should preferably be able to be replaced by a "fresh" s "home" for recharging and maintenance. Nice if this could be handled autono system.

https://www.wfp.org/stories/drone-worship-why-skys-limit-humanitarian-wifi

BDGS DE Architectu... | TOOLS BDGS | SAN ADDM Discov... | E2C Karlskrona | WFP | Motorola program... | Radio Mobile WEB... | ER | Humanitarian Booki... | UN UNDSS

WFP World Food Programme

SAVING LIVES CHANGING LIVES

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Home › Stories › Drone worship: Why sky's the limit for humanitarian WiFi

Drone worship: Why sky's the limit for humanitarian WiFi

How the World Food Programme could change how connectivity is brought to emergencies

29 December 2021, Peyvand Khorsandi

Innovation and technology



WFP's Patrick McKay, right, check the WiFi tub as Peter Opdam, CEO and founder of UK-based developer Clogworks Technologies looks on.

Drone worship: Why sky's the limit for humanitarian WiFi | World Food Programme (wfp.org)

Highlight of missions 2000 – 2021

Impact

2000 – Extreme floods in Algeria & severe drought in Tajikistan
2001 – Flooding in Hungary and earthquakes in Pakistan, Peru, El Salvador. First emergency response on large scale to Kabul, Afghanistan
2002 – Rail disaster in Tanzania and food crisis in Lesotho. Humanitarian Aid to Afghanistan.
2003 – Earthquakes strike Algeria and Iran. Hurricane in Caribbean. Aid to Liberia.
2004 – Tsunami devastates Indonesia and Sri Lanka.
2005 – Hurricane Katrina hits the US. Earthquakes in Pakistan.
2006 – ICT support for aid workers in South Sudan. Earthquake victims in Pakistan.
2007 – Establish connectivity for aid worker in Central African Republic and in Peruvian Earthquake.
2008 – Establish connectivity for Save the Children in South Sudan.
2009 – Assessment of ICT needs in Philippines. ICT support in demining mission in DRC.
2010 – Earthquake strikes Haiti.
2011 – "One UN" initiative.
2012 – ICT support in South Sudan, Mozambique, Tanzania and Uganda.
2013 – Super typhoon Haiyan hits the Philippines.
2014 – West Africa during the Ebola crisis. Continuous support in South Sudan.
2015 – Earthquakes in Nepal.



2016 – Hurricane Matthew struck Haiti. Mission in South Sudan ended.
2017 – Hurricane Irma and Maria, relief efforts in the Dominica, Caribbean.
2018 – WFP IT support in Libya
2019 – Cyclone Idai, Mozambique.
2019 – Refugee crisis in Bangladesh – Cox's Bazar
2019 – Hurricane Dorian – Bahamas
2019 – Venezualian Refugees – Colombia
2020 – Explosion in Beirut port – Lebanon
2020 – **20 Years Annivresary Ericsson Response**
2021 – Ending Bangladesh support– Cox's Bazar
2021 – Fooding in western Germany– Germany
2021 – Earthquake – Haiti
2021 – Connectivity for Refugee registration – Colombia
2022 – Typhoon Odette – Philippines
2022 – Refugee support – Moldova
2022 – Refugee support – Colombia

Türkiye 2023

— ETC Earthquake Response February/March mainly Gaziantep, Hatay province.

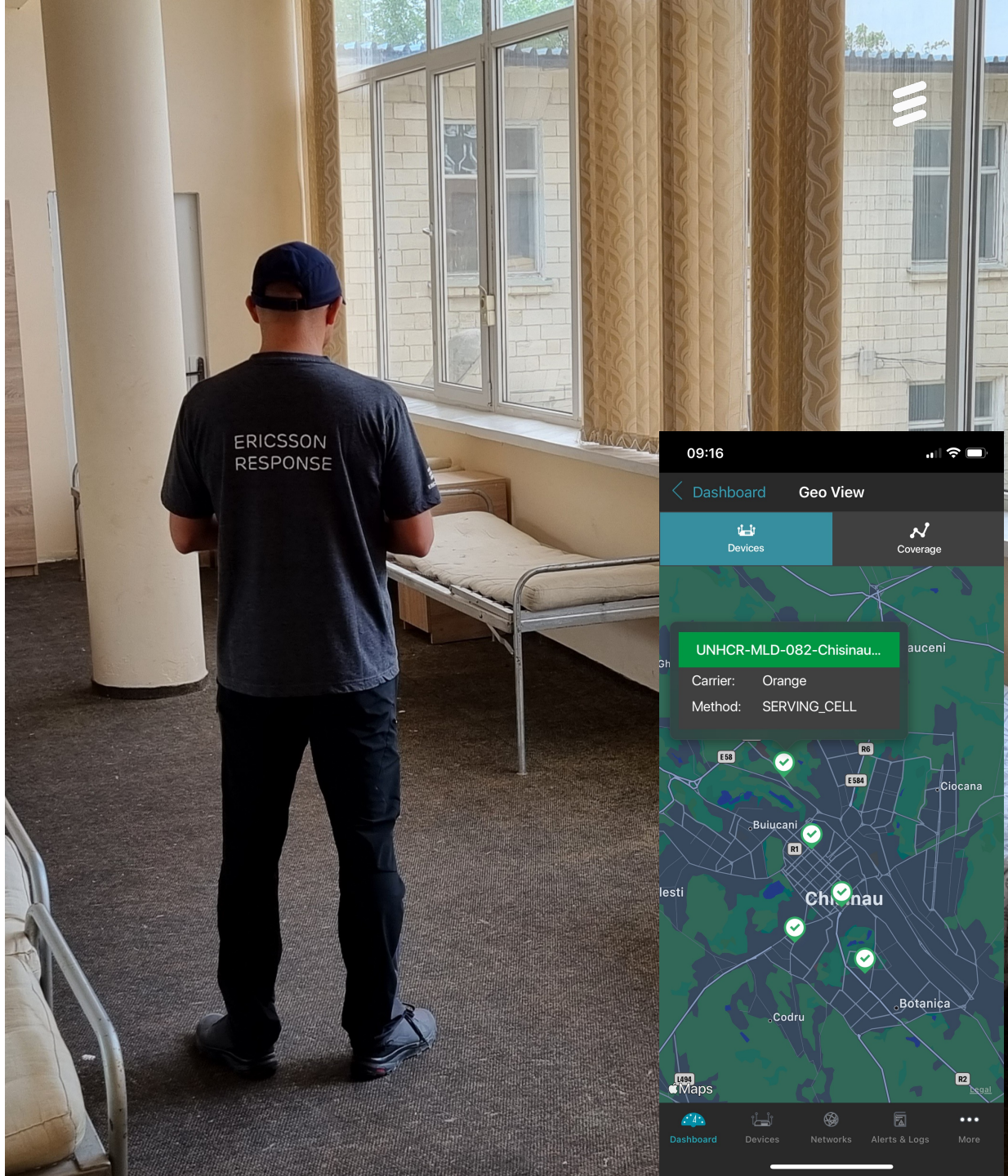
— Charging station units are operational in two temporary settlements in Hatay. Affected populations and the Ministry of Education (which is managing the settlement) are using the service to power communications devices for improved access to critical communications services and information

[Türkiye | Emergency Telecommunications Cluster (ETC) (etcluster.org)]



Moldova 2022

- Right after the Ukraine war started Feb 24th, refugees started to leave the war-torn country
- Connectivity to over 35 location cooperation with RETS partners, supporting services to refugees, like Blue Dots, transit points and accommodation centers



Colombia 2021/2022

- Government of Colombia offered Temporary Protection Status (TPS) for refugees from Venezuela, giving access to healthcare, education and job market
- 170 sites were connected by RETS and partners where 2.4 million refugees registered for TPS

- In 2021 Ericsson Response and RETS partner implemented a community connectivity project
  - 7 points of internet access
  - 1200 people got access to internet





Bangladesh 2019

- The Rohingya refugee crises, in August 2017 more than 773,000 fled the Rakhine State, Myanmar.

- Ericsson Response Volunteers were embedded in the WFP response for almost a year, proving connectivity for the WFP SCOP (web-based beneficiary management platform)

- October 2022, 5 years later approximately 1 million stateless Rohingyas still live in the 34 extremely congested camps.

SCOPE: wfp272586.pdf

Caribbean 2017

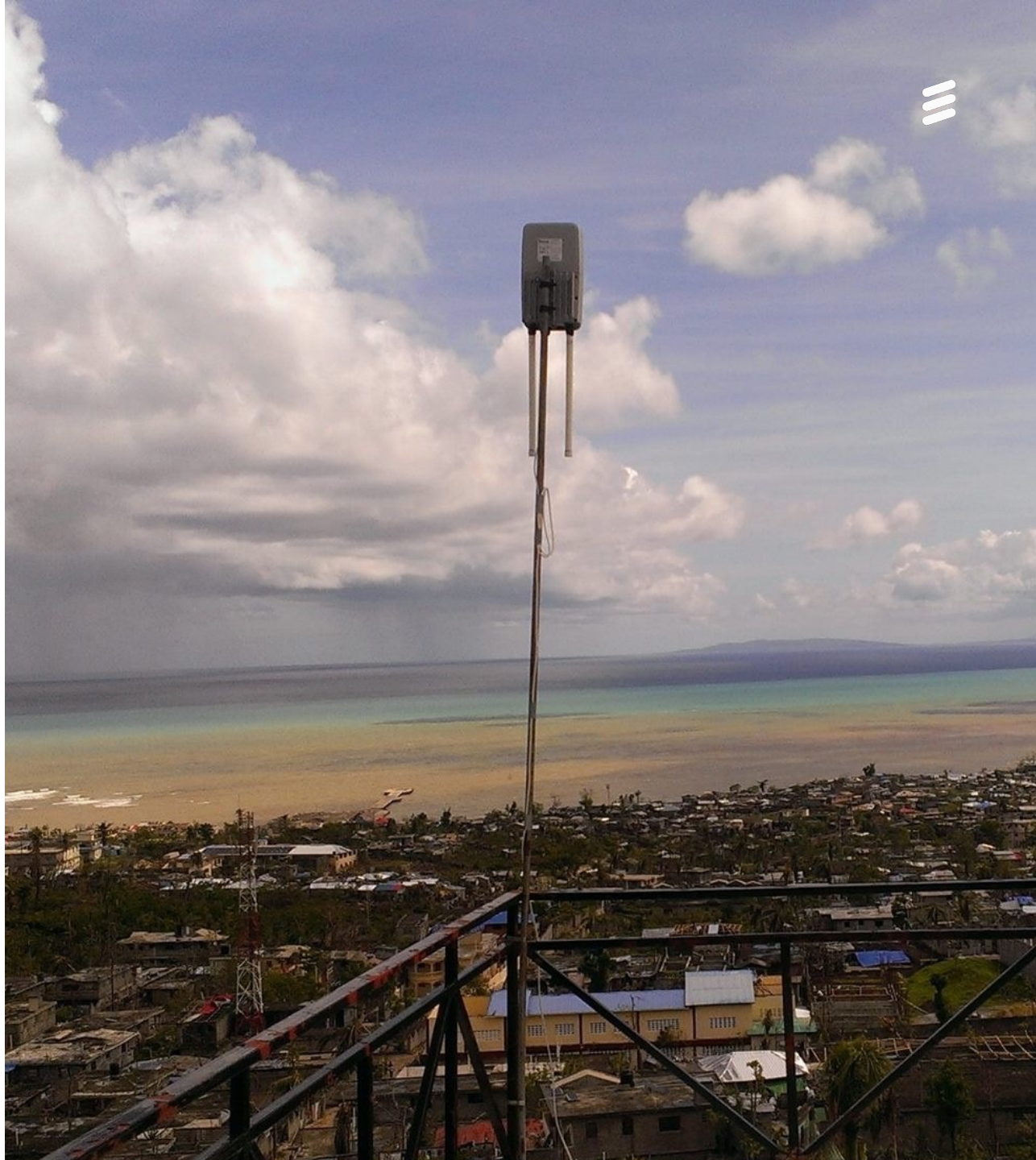
- Connectivity to over 2900 users (peak data) from over 50 relief organizations in cooperation with ETC partners, including limited access for the local population
- Covers both major airports in the country, as well as health care facilities and the main port
- GSMA Global Mobile Award for Best Mobile Innovation supporting Emergency or Humanitarian Situations at the 2018 Mobile World Congress

Case film



Haiti 2016

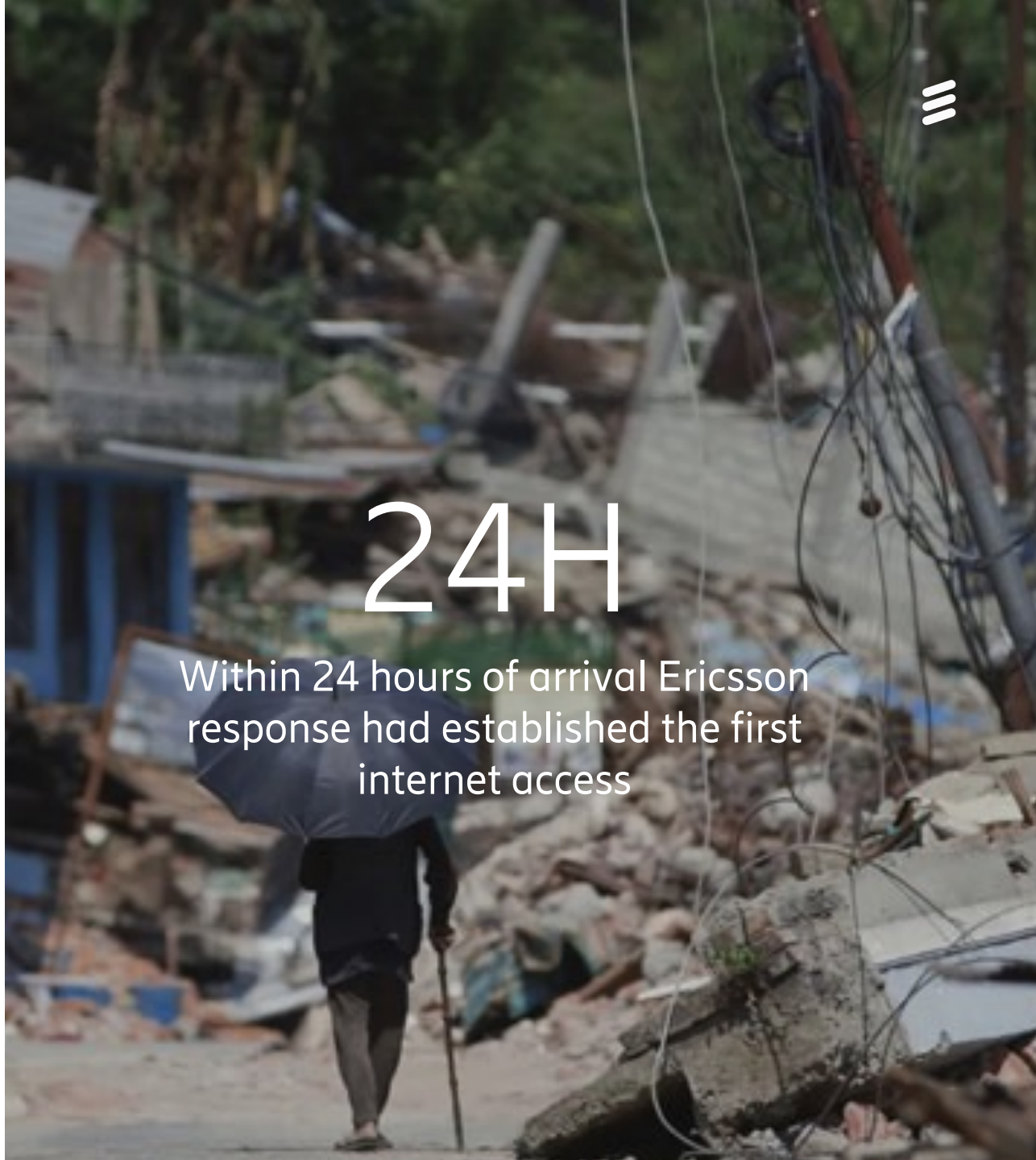
- In 2016, Hurricane Matthew struck Haiti
- Ericsson Response was deployed under the ICT Humanitarian Response Working Group for Haiti
- 19 sites were established
  - Supporting over 900 humanitarian users and 30 agencies



Nepal 2015

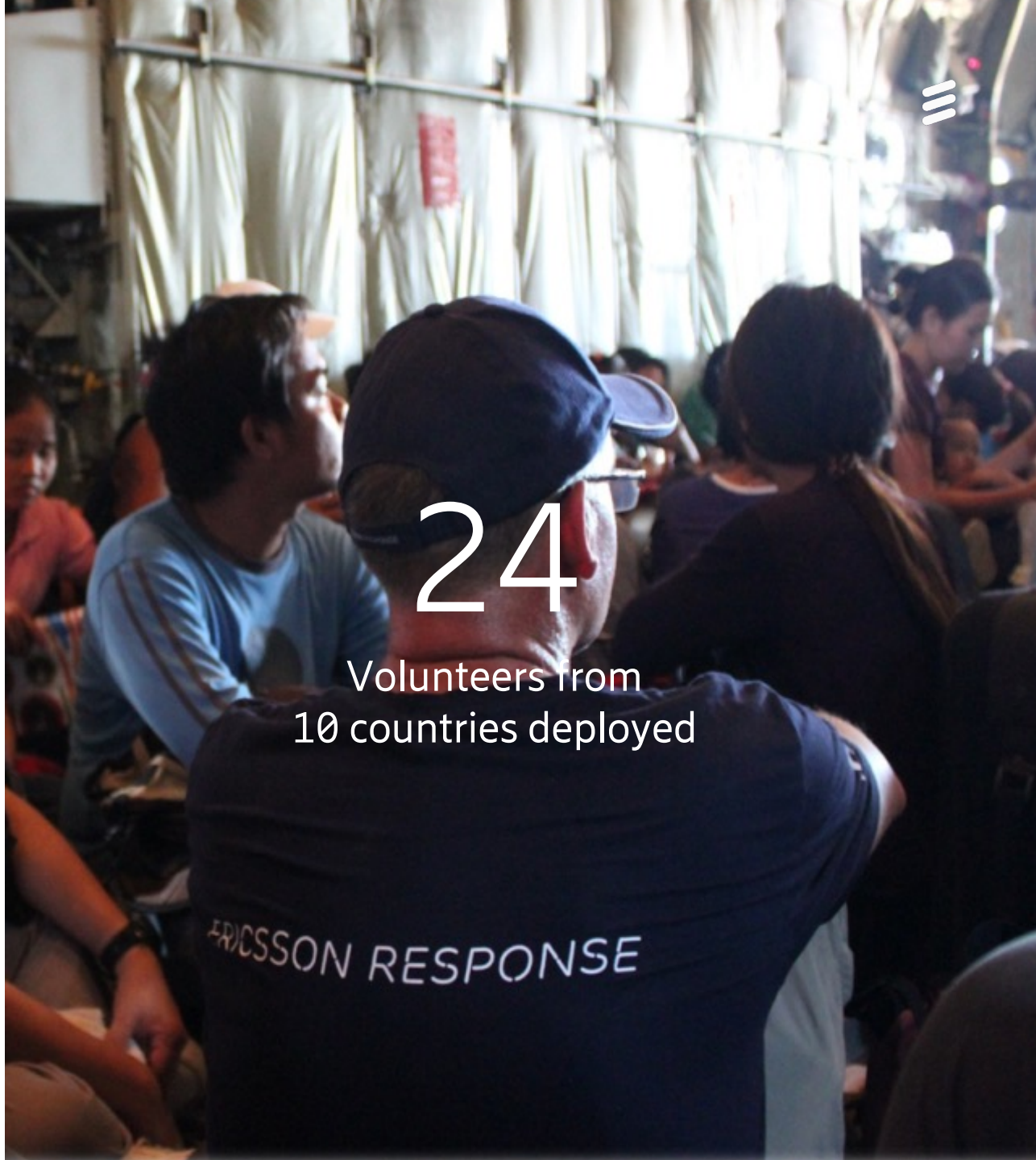
— On April 25, 2015 a devastating earthquake struck Nepal
— Ericsson Response as part of the Emergency Telecommunications Cluster was immediately deployed to assist in the relief efforts
— During the first month, supporting more than 1000 users from over 130 humanitarian organizations aiding the millions of people affected by the disaster

Case film



Philippines 2013

- 6 months presence in Tacloban, Guiuan and Roxas
- Working with partners WFP, UN OCHA
- 3 WIDER (WLAN) kits deployed
- Services provided across 12 locations to 6 400+ humanitarian workers from 40 different organizations
- More than 35 000 metric tons of relief items moved by land, sea and air
- 2.8 million people reached with food assistance by WFP



S4C/Service for communities

- Connecting Communities
- Supplying connectivity not only to UN, NGO's and Governmental organizations.
- The backhaul has previously always been the bottleneck in a deployment with geostationary satellite, but with the introduction of LEO (Low-Earth Orbit) Satellite and more stable ISP's and NMO (network mobile operators) the backhaul can be quite good.

Case Film





https://www.ericsson.com